ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Section A: objective questions (30) · Section B: microcontroller programming structure, digital and analogue input/output, decisions and loops, functions, serial communication and debugging (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Code in this paper is written for an Arduino Uno (C/C++). Minor syntax errors will not be penalised if the logic is correct.
5. Write neatly and legibly. Indent your code clearly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The function that runs only once when the Arduino starts is …

A loop(). B main(). C setup(). D start().

1.1.2 Which function reads the value of an analogue input pin?

A analogRead() B digitalRead()
C analogWrite() D pinMode()

1.1.3 The range of values returned by analogRead() on an Arduino Uno is …

A 0 to 255. B 0 to 5. C 0 to 100. D 0 to 1 023.

1.1.4 analogWrite() on a PWM pin accepts values from …

A 0 to 1 023. B 0 to 255. C 0 to 5. D 1 to 100.

1.1.5 Which statement sets pin 7 as an input with the internal pull-up resistor switched on?

A pinMode(7, OUTPUT); B digitalWrite(7, INPUT);
C analogRead(7, HIGH); D pinMode(7, INPUT_PULLUP);

1.1.6 The data type that stores a decimal number such as 3.75 is …

A int. B float. C boolean. D char.

1.1.7 delay(2000) pauses the program for …

A 2 000 seconds. B 0,2 seconds. C 2 seconds. D 20 seconds.

1.1.8 The operator that means "logical AND" in C/C++ is …

A && B || C ! D ==

1.1.9 Serial.begin(9600) sets the …

A pin number. B communication speed (baud rate).
C delay time. D voltage.

1.1.10 Every statement in C/C++ ends with a …

A colon (:). B full stop (.). C comma (,). D semicolon (;).

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 C/C++ is case-sensitive, so Delay() and delay() are different.

1.2.2 A variable declared inside loop() keeps its value from one loop to the next unless it is declared static.

1.2.3 digitalWrite(13, HIGH) sets pin 13 to about 5 V.

1.2.4 The = operator compares two values.

1.2.5 A for loop is suitable to repeat an action an exact number of times.

1.2.6 millis() returns the number of milliseconds since the board started running.

1.2.7 An Arduino output pin can drive a 12 V motor directly.

1.2.8 Comments in C/C++ can start with //.

1.2.9 The map() function can change a value from one range to another.

1.2.10 The Serial Monitor can be used to display variable values while debugging.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Turns source code into machine code for the microcontroller	A	switch statement
1.3.2	Ready-made code that adds functions, e.g. Servo.h	B	library
1.3.3	A name for a value that never changes, e.g. const int LED = 13;	C	compiler
1.3.4	Block of code with a name that can be called many times	D	argument
1.3.5	Value sent into a function when it is called	E	serial communication
1.3.6	Several values of the same type stored under one name	F	constant
1.3.7	Selects one of many cases depending on the value of a variable	G	function
1.3.8	Variable that can be used in every function	H	uploading
1.3.9	Sending data one bit at a time between the Arduino and a computer	I	global variable
1.3.10	Sending the compiled program to the board	J	array

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: PROGRAM STRUCTURE AND DIGITAL INPUT/OUTPUT

2.1 Explain the purpose of setup() and loop() in an Arduino program. (4)

2.2 Study the code below and answer the questions.

```
const int button = 2;
const int led = 9;
void setup() {
  pinMode(button, INPUT_PULLUP);
  pinMode(led, OUTPUT);
}
void loop() {
  if (digitalRead(button) == LOW) {
    digitalWrite(led, HIGH);
  } else {
    digitalWrite(led, LOW);
  }
}
```

2.2.1 Why does the program test for LOW and not HIGH when the button is pressed? (2)

2.2.2 Describe what the program does. (2)

2.2.3 Change the program so that each press of the button toggles the LED (on, then off on the next press). (6)

2.3 Write a program that makes three LEDs on pins 4, 5 and 6 light up one after the other like a running light, each for 300 ms, repeating forever. Use a for loop. (6)

[20]

QUESTION 3: ANALOGUE INPUT AND OUTPUT

3.1 A potentiometer is connected to pin A0 and an LED to PWM pin 10. Write a program that sets the brightness of the LED according to the position of the potentiometer. Use map(). (6)

3.2 A temperature sensor gives 10 mV per °C (0 V at 0 °C) and is connected to A1.

3.2.1 Write the statements to read the sensor and convert the reading to degrees Celsius. (4)

3.2.2 Calculate the temperature if analogRead() returns 61. (3)

3.3 Explain what PWM is and why analogWrite() does not give a true analogue voltage. (3)

[16]

QUESTION 4: DECISIONS, LOOPS AND FUNCTIONS

4.1 A robot measures the distance to an obstacle in centimetres (variable dist). Write an if–else if–else structure that: drives forward if dist is more than 30; slows down if dist is from 15 to 30; stops and turns if dist is less than 15. Use the functions forward(), slow() and turnAround(). (6)

4.2 Write a function called average that takes an array of five int readings and returns their average as a float. (6)

4.3 Explain the difference between using delay() and using millis() for timing in a robot program. (4)

[16]

QUESTION 5: SERIAL COMMUNICATION AND DEBUGGING

5.1 Write the statements to start serial communication at 9600 baud and display the message "Distance: " followed by the value of the variable dist and "cm" on one line. (4)

5.2 The code below should flash an LED five times, but it contains errors. Identify FOUR errors and correct them.

```
1  int led = 13
2  void setup() {
3    pinMode(led, INPUT);
4  }
5  void loop() {
6    for (int i = 0; i <= 5; i++) {
7      digitalwrite(led, HIGH);
8      delay(500);
9      digitalWrite(led, LOW);
10   }
11   while (true);
12 }
```

(8)

5.3 Give THREE good programming practices for writing code that is easy to read and maintain. (3)

5.4 Explain why you should test a robot program in small parts. (3)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER